Set Operators in SQL Server

1. What are Set Operators?

Set Operators are used to **combine the results of two or more SELECT statements** into a single result set.

They work based on **set theory in mathematics**.

SQL Server Set Operators

1. **UNION**
2. **UNION ALL**
3. **INTERSECT**
4. **EXCEPT**

Important Rules for Set Operators

Before using set operators, the following conditions must be satisfied:

1. **Number of columns must be the same**
2. **Data types must be compatible**
3. **Column order must be the same**
4. **Only one ORDER BY clause allowed at the end**

Example

✓ Correct

```
SELECT Id, Name FROM Table1

UNION

SELECT Id, Name FROM Table2
```

❌ Incorrect

```
SELECT Id, Name FROM Table1

UNION
```

SELECT Id FROM Table2

(Number of columns mismatch)

Example Tables (Student Scenario)

Table: Students2023

StudentID	Name	Course
1	Ravi	SQL
2	Priya	Python
3	Amit	Java

Table: Students2024

StudentID	Name	Course
3	Amit	Java
4	Sneha	.NET
5	Rahul	SQL

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1. UNION Operator

Definition

UNION combines results from two queries and removes duplicate rows.

Syntax

SELECT column_list FROM table1

UNION

SELECT column_list FROM table2

Example

SELECT Name FROM Students2023

UNION

SELECT Name FROM Students2024

Result

Name
Ravi
Priya
Amit
Sneha
Rahul

Notice:

- **Amit appears only once**
- UNION removes duplicates automatically.

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2. UNION ALL

Definition

UNION ALL combines results but keeps duplicates.

Syntax

SELECT column_list FROM table1

UNION ALL

SELECT column_list FROM table2

Example

SELECT Name FROM Students2023

UNION ALL

SELECT Name FROM Students2024

Result

Name

Ravi

Priya

Amit

Amit

Sneha

Rahul

Here:

- **Amit appears twice**
- Because UNION ALL **does not remove duplicates**

UNION vs UNION ALL (Very Important Interview Question)

Feature	**UNION**	**UNION ALL**
Removes duplicates	Yes	No
Performance	Slower	Faster
Sorting required	Yes	No
Usage	Unique results	All records

Why UNION is slower?

Because SQL Server internally performs:

SORT + DISTINCT

to remove duplicates.

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3. INTERSECT Operator

Definition

INTERSECT returns only the common rows between two queries.

Syntax

SELECT column_list FROM table1

INTERSECT

SELECT column_list FROM table2

Example

SELECT Name FROM Students2023

INTERSECT

SELECT Name FROM Students2024

Result

Name
Amit

Only **Amit exists in both tables**.

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4. EXCEPT Operator

Definition

EXCEPT returns rows from the first query that are NOT present in the second query.

Syntax

SELECT column_list FROM table1

EXCEPT

SELECT column_list FROM table2

Example

SELECT Name FROM Students2023

EXCEPT

SELECT Name FROM Students2024

Result

Name

Ravi

Priya

Because these students exist **only in Students2023**.

Visual Understanding

Assume two sets:

Table A = {1,2,3,4}

Table B = {3,4,5,6}

UNION

{1,2,3,4,5,6}

UNION ALL

{1,2,3,4,3,4,5,6}

INTERSECT

{3,4}

EXCEPT

{1,2}

Practical Example (Employee Tables)

Table: Employees_Hyd

EmpID	**Name**
1	Ravi
2	Priya
3	John

Table: Employees_Bangalore

EmpID	Name
3	John
4	Meena
5	Rahul

UNION Example

SELECT Name FROM Employees_Hyd

UNION

SELECT Name FROM Employees_Bangalore

Result

Ravi

Priya

John

Meena

Rahul

INTERSECT Example

SELECT Name FROM Employees_Hyd

INTERSECT

SELECT Name FROM Employees_Bangalore

Result

John

EXCEPT Example

SELECT Name FROM Employees_Hyd

EXCEPT

SELECT Name FROM Employees_Bangalore

Result

Ravi

Priya

Using ORDER BY with Set Operators

Only one **ORDER BY** allowed at the end.

SELECT Name FROM Students2023

UNION

SELECT Name FROM Students2024

ORDER BY Name